



ARP Biotech

# Sample Sequencing and Data Analysis Report

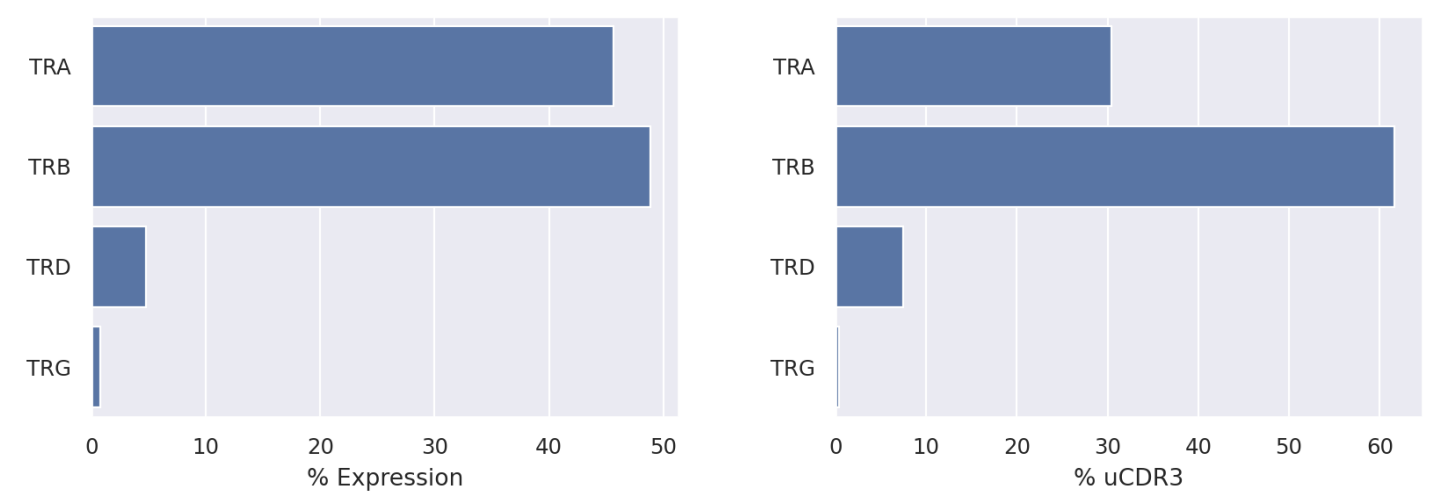
## SD\_01\_MCT0209

### Data Analysis QC Info

| Sample ID     | RNA Conc. (ng/uL) | RNA 260/280 | Decoded Reads | High Quality Reads | UMIs    | Reads per UMI | Total Receptor RNA |
|---------------|-------------------|-------------|---------------|--------------------|---------|---------------|--------------------|
| SD_01_MCT0209 | 10.00             | 1.95        | 6,697,586     | 6,040,242          | 705,168 | 8.57          | 675,970            |

### T cell Expression and Unique CDR3 Counts

| Index              | TRA    | TRB    | TRD   | TRG   |
|--------------------|--------|--------|-------|-------|
| Expression %       | 45.61% | 48.88% | 4.77% | 0.74% |
| Total Receptor RNA | 256852 | 275309 | 26838 | 4184  |
| Unique CDR3 %      | 30.45% | 61.65% | 7.47% | 0.43% |
| Total Unique CDR3s | 97939  | 198302 | 24034 | 1380  |



### T cell Diversity Indexes

All diversity indexes are calculated using the CDR3 peptide count distribution. d50 = % of unique CDR3s that make up the top 50% of the total reads. Note: Diversity indexes are not comparable across chain.

| Index             | TRA   | TRB   | TRD   | TRG   |
|-------------------|-------|-------|-------|-------|
| d50               | 9.81  | 30.58 | 44.17 | 4.35  |
| Shannon entropy   | 10.72 | 11.96 | 10.01 | 5.95  |
| Pielou's evenness | 0.93  | 0.98  | 0.99  | 0.82  |
| Gini index        | 0.53  | 0.25  | 0.10  | 0.61  |
| Largest clone %   | 0.86% | 0.02% | 0.16% | 7.98% |

## Treemaps

Each rectangle represents a unique clone (CDR3 peptide + V gene + J gene) with its size proportional to its expression.

TRA



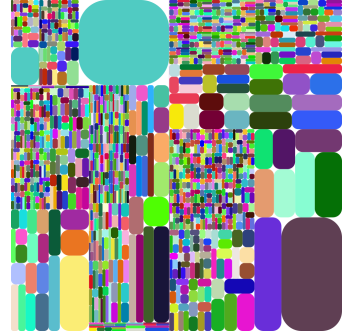
TRB



TRD



TRG



## T cell Treemap

Each outer rectangle represents a chain, with its size representative of its relative expression out of the 4 T cell chains (productive sequences). These treemaps are always read from left to right, then top to bottom in this order: TRA, TRB, TRD, TRG.





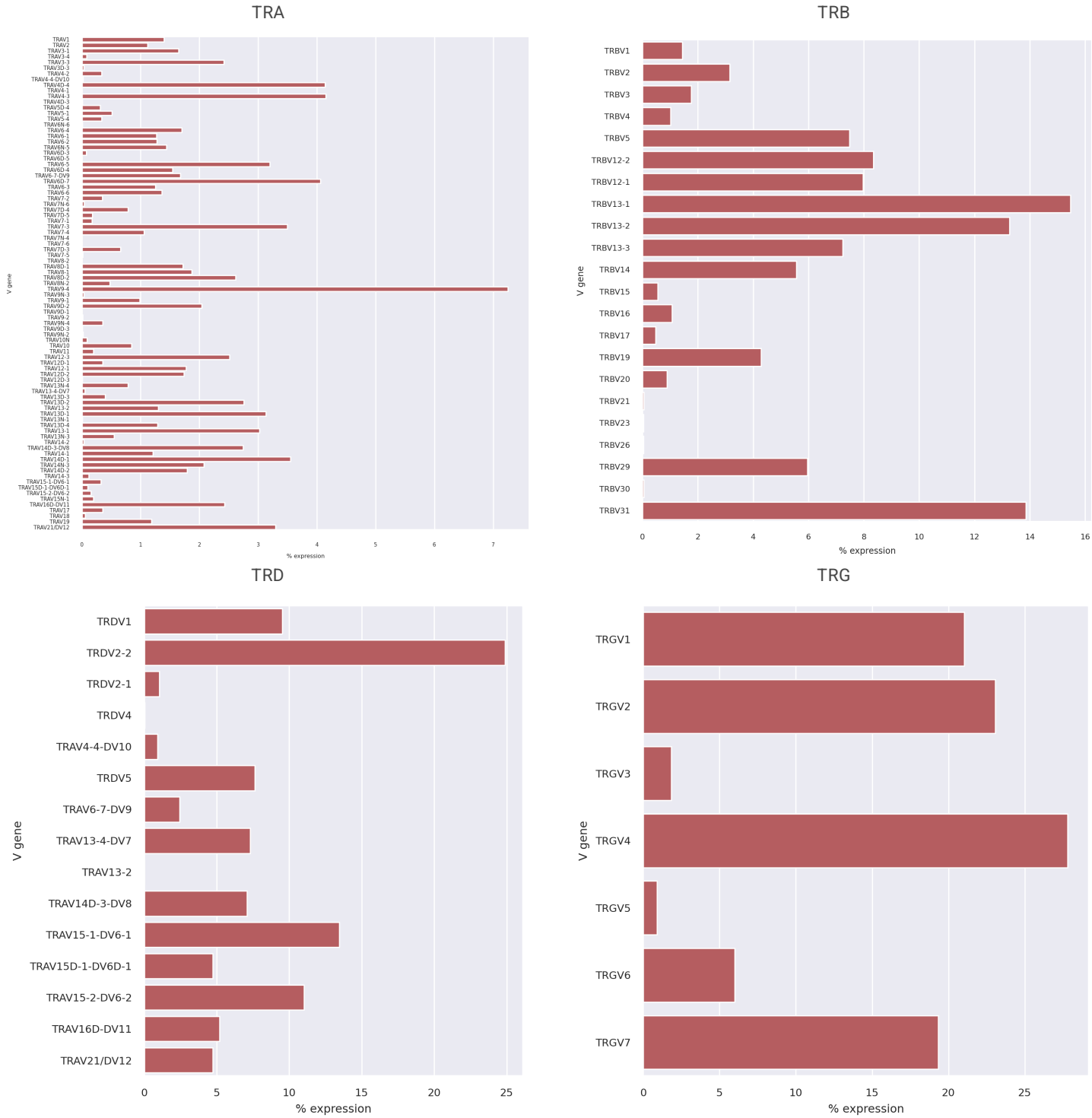
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### T cell V Gene Usage

Percentage expression of each V gene.





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### T cell J Gene Usage

Percentage expression of each J gene.

